Lab Solution: Advanced ETL

Run the following cell to setup the lab environment

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```
%run ../Includes/Setup-Lab
```

Q1- Interacting with JSON data

Review the nested data structures of the **profile** column in the **students** table created in the previous lab

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```
SELECT email, profile
FROM students
```

7

```
DESCRIBE students
```

Use the appropriate syntax to access the **last_name** and **city** information from the **profile** column

9

```
-- ANSWER
SELECT email, profile:last_name AS student_surname, profile:address:city AS student_city
FROM students
```

Q2- Higher Order Functions

Review the array column **courses** in the **enrollments** table created in the previous lab

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```
SELECT enroll_id, courses
FROM enrollments
```

Filter this array column to keep only course elements having subtotal greater than 40

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```
-- ANSWER
SELECT
  enroll_id,
  courses,
  FILTER (courses, i -> i.subtotal > 40) AS large_totals
FROM enrollments
```

Q3- SQL UDFs

Define a UDF function named **get_letter_grade** that takes one parameter named **gpa** of type DOUBLE. It returns the corresponding letter grade as indicated in the following table:

GPA (4.0 Scale)	Grade Letter
3.50 - 4.0	A
2.75 - 3.44	B
2.0 - 2.74	C
0.0 - 1.99	F

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```
-- ANSWER
CREATE FUNCTION get_letter_grade(gpa DOUBLE)
RETURNS STRING
RETURN CASE
          WHEN gpa >= 3.5 THEN "A"
          WHEN gpa >= 2.75 AND gpa < 3.5 THEN "B"
          WHEN gpa >= 2 AND gpa < 2.75 THEN "C"
          ELSE "F"
       END
```

Let's apply the above UDF on the **students** table created in the previous lab

Fill in the below query to call the defined UDF on the **gpa** column

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```
-- ANSWER
SELECT student_id, gpa, get_letter_grade(gpa) as letter_grade
FROM students
```

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